

10

UNSUSTAINABLE HUMAN–ECOSYSTEM INTERACTION

Past and present experiences with human–ecosystem interaction can provide lessons about how to avoid mistakes. Environmental problems are not entirely new. Although most societies in the past have lived in harmony with the environment most of the time, there have been occasions when particular societies have had very **unsustainable** interactions with the environment. Considering the consequences, it is natural to ask: ‘How could people make such serious mistakes in the past, and why does modern society continue to repeat such mistakes today?’

As a rule, human–ecosystem interaction is sustainable when social system and ecosystem are coadapted. Conversely, interaction is less sustainable when coadaptation is weak. Sudden changes in the social system or ecosystem can disrupt coadaptation, setting in motion a chain of effects that reduces an ecosystem’s ability to provide essential services. This chapter will illustrate how coadaptation can be lost when people migrate to new places with completely different ecosystems – ecosystems with which they have no previous experience. It will also describe how coadaptation can decline after sudden social system changes such as new technologies.

This chapter will then turn to powerful social forces that cause unsustainable interaction between modern social systems and ecosystems. The basic source of unsustainable human–ecosystem interaction today is an expanding human population, coupled with an expanding economy that makes excessive demands on ecosystems. The chapter will describe how modern economic institutions motivate individuals to use ecosystem resources in ways that are unsustainable. It will describe the role of urbanization, which erodes social system–ecosystem coadaptation as urban populations become alienated from their environmental support system. The rise and fall of past civilizations provides

insights into urbanization and economic development that are proceeding on a global scale today. The chapter will show how aggressive commercial exploitation of ecosystem resources can lead to wishful thinking about the intensity of resource use that an ecosystem can sustain. It will close with the precautionary principle as a prudent way to ensure sustainable resource use in the face of incomplete knowledge about how much resource depletion ecosystems can sustain.

HUMAN MIGRATIONS

human migrations

Unsustainable interaction between people and ecosystems has often been associated with human migrations. When people move to a new area where the ecosystem is different, they typically have little knowledge about the new ecosystem and lack appropriate **social institutions** and technology for sustainable interaction. This appears to have occurred when the first human inhabitants of North America migrated there from Asia about 13,000 years ago. When the people arrived, North America had numerous species of large mammals similar to the impressive fauna in East Africa today. Most species of large mammals disappeared within a few centuries of their arrival, probably due to overhunting. We do not know that the Native Americans were definitely responsible, but it appears likely. Many large animal species in Europe and Australia also disappeared soon after the first humans migrated to those continents.

During the centuries that followed, Native American social systems coevolved with their local ecosystems until the social systems and ecosystems were generally coadapted. While the cultures of different tribes and the details of their interaction with the environment were diverse, social institutions for sustainable interaction with the ecosystem were a common part of Native American cultures. Tribal territoriality was important for defining clear ownership of common property resources, such as deer and other animals that the Native Americans hunted on a sustainable basis.

Coadaptation did not mean that Native Americans left the environment in a completely natural state. In fact, they modified their ecosystem in many ways. They used fire to create small patches with early stages of ecological succession such as grass meadows in parts of North America that were mainly climax forest. A diverse mixture of different stages of ecological succession

human
migrations

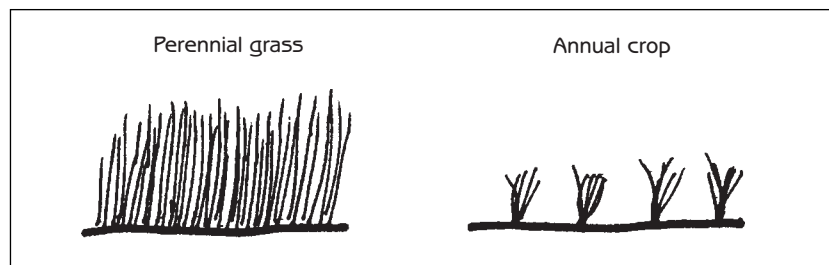
created a landscape mosaic with more favourable hunting and a greater variety of other ecosystem 'services' than was possible from one kind of ecosystem alone.

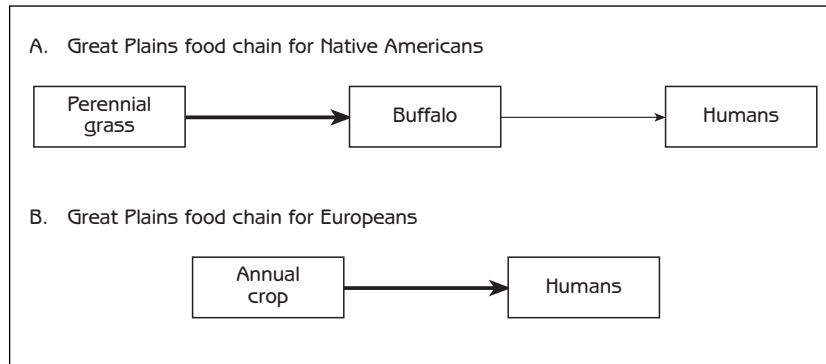
The Great Plains of North America had a deep, rich topsoil and tall, dense perennial grasses that provided food for large herds of buffalo. The perennial grasses were a mixture of native species, a natural polyculture adapted to the windy conditions of the Great Plains. Because they were perennial, the grasses covered the soil completely throughout the year, protecting it from wind erosion (see Figure 10.1). Native Americans adapted to the Great Plains ecosystem by using buffalo as their main resource (see Figure 10.2a). Because their religion emphasized respect for nature, wild animals could only be killed when required for food or other basic needs. They used almost every part of the buffalo's body for food, clothing or building materials for their houses.

When Europeans invaded North America about 300 years ago, they exploited North American resources in an unsustainable manner because they did not have the values, knowledge, technology and other social institutions appropriate for sustainable interaction with the North American ecosystem. They perceived the vast resources of the continent as virtually unlimited, and they considered the transformation of natural ecosystems to agricultural and urban ecosystems of European design to be unequivocal progress. They considered the Native American social system to be a primitive stage of human social evolution inappropriate for modern times. Many Europeans considered the Native Americans themselves to be an inferior race destined for extinction.

When European immigrants reached the Great Plains, they saw buffalo as a source of money. Professional hunters killed buffalo by the millions, selling buffalo hides to the international leather market. Within 20 years the buffalo were reduced from a population of 60 million to almost nothing. The Native Americans living on the Plains were reduced to starvation and desperation when the buffalo – their main source of food – was destroyed and large numbers of Europeans settled on their land to farm it. The

Figure 10.1
Comparison of
the natural Great
Plains ecosystem
(perennial
grasses) with
annual crops
planted by
Europeans



**Figure 10.2**

Great Plains food chain for people before and after the European invasion of North America

human
migrations

Native Americans responded with war, but they lost the war and their land, and were subsequently reduced to a marginal existence.

The European farmers who replaced the Native Americans on the Great Plains grew **monocultures** of wheat, corn and other **annual crops** (see Figure 10.2b). These crops provided a shorter food chain than grass and buffalo, so the farmers were able to capture a larger percentage of the Great Plains biological production than the Native Americans had obtained from buffalo. However, unlike the natural vegetation of the Great Plains, these crops were not adapted to protect the soil from wind erosion. Annual crops are food plants that have a new generation each year. They do not cover the soil completely like perennial grass, and because annual crops are on the fields for only part of each year, the soil is not protected during the other part of the year (see Figure 10.1). This kind of farming worked well under European conditions, where wind erosion is not such a serious problem, but it could not protect the soil under weather conditions on the Great Plains. As a consequence, most of the Great Plains topsoil has been carried away by wind since Europeans began farming there 120 years ago. The soil has lost its natural fertility and now provides high crop yields only with large fertilizer inputs. The latest chapter in the Great Plains story comes from a small number of scientists who are developing new agricultural ecosystems with **polycultures** of native perennial grasses that produce enough grain to be of commercial use. Agricultural ecosystems that mimic the natural ecosystems of the Great Plains should reduce erosion because they cover the soil better.

Migrations continue to be important around the world as millions of land-hungry people move from overpopulated areas to areas with fewer people. Governments often encourage these migrations, a policy that is not ecologically wise when the population movement is to an area that has fewer people because the human carrying capacity cannot be changed readily by modern

140 *Human Ecology***human
migrations**

technology. Immigrants usually damage the environment and reduce the carrying capacity, not only because their large numbers force them to overexploit local resources but also because their cultural traditions do not provide the worldview, values, knowledge, technology and social institutions that are needed for sustainable interaction with their new environment.

Millions of rural people have migrated from overpopulated Asian lowlands to less crowded mountain regions in countries such as Vietnam and the Philippines during recent years. Relatively few people lived in the mountains in the past because the human carrying capacity of mountains is less than river valleys and coastal plains which have deep, flat, fertile soil. Mountain people have farmed the steep mountain land for centuries without environmental damage because their agriculture is coadapted with the mountain ecosystem. Lowland people who move to the mountains often use lowland agricultural methods that are not sustainable in the mountains because their agriculture is not designed to protect steep hillsides from erosion.

The same thing is happening with human migrations to tropical rainforests. Millions of Brazilians have moved to the Amazon, and millions of Indonesians have moved from the overpopulated island of Java to farm Indonesia's outer islands, where lush green forests have thrived for thousands of years on some of the world's poorest soils. Rainforest ecosystems maintain the fertility of their soil with complex adaptations that prevent the loss of scarce mineral nutrients from the forest ecosystem. Until recently, only a small number of people lived in rainforests, and they used the forest in ways that did not interfere with the sustainability of the ecosystem, such as hunting and gathering and slash-burn agriculture.

Today a large number of people are cutting rainforests and replacing forest ecosystems with farms that are unsustainable on nutrient-poor rainforest soils. Their agricultural ecosystems lack the intricate mechanisms that allow rainforest ecosystems and the traditional agriculture of the region to maintain soil fertility. These inappropriate agricultural ecosystems stop producing within a few years. The land may then be used for grazing, producing beef for export to industrialized nations (the hamburger connection). Eventually even grasses may not grow, or there is grazing-induced succession to grass too tough for cattle to eat, and the land is subsequently abandoned. It is a 'tropical desert' with soil so severely damaged that it can be many years before it will once again support a rainforest or human use. Tropical forest immigrants then move to new places, where they clear-cut forests in order to farm soil that has not yet lost its

fertility. Eventually, immigrant farmers will cut down all the forest and, in the end, these people will still not have a suitable place in which to live.

The story of human migrations shows us how human–ecosystem interaction can change with the passage of time. Migrating people bring a social system that is not adapted to their new environment, but with time they have the potential to reorganize and adjust their social systems to the new conditions. Problems of unsustainable activities by migrants will become increasingly common as more people in the developing world move from overpopulated areas to areas that are less populated but also less suited to accommodate a population increase. The most important lesson from this example is that people can learn and adapt. International and national policies for sustainable development need to assist migrant people to learn from people who have lived in the same area for many generations, so that migrants can adapt quickly to their new environment, doing as little damage as possible.

new
technologies

NEW TECHNOLOGIES

People often cause extensive environmental damage when they adopt a new technology. They do not know the environmental consequences of the new technology, and their social system does not have the institutions to use the technology in ways that are environmentally sustainable. For example, traditional hunting societies used weapons such as spears, bows and arrows, or blowguns with poison darts, which were not effective enough to damage the populations of their food sources. It was safe for hunters to kill as many animals as possible. However, the same resources can be overexploited when a new technology, such as guns, is introduced and hunters continue to kill as many animals as possible. The hunters' knowledge of nature may be immense, but their culture may not have evolved a conservation ethic if it was unnecessary in the past. Fishermen around the world now use monofilament nylon nets, which are much more effective than traditional fishing nets because fish cannot see them in the water. The result has been severe overfishing and decline of fish populations in many parts of the world.

Market changes can disrupt the sustainable use of natural resources because new market opportunities encourage people to use production technologies with which they have little

**portable
capital**

previous experience. For example, the rapid growth of developing world cities in recent years has created expanding markets for European crops such as cabbage, stimulating large-scale commercial production of these crops in mountains where they were not cultivated in the past. Tropical mountain hillsides are very susceptible to erosion. If they are not protected from rainfall by plants covering the soil, rainwater can carry away hundreds of tonnes of soil from each hectare of hillside every year. Traditional mountain agricultural ecosystems have been sustainable for centuries because they use crops that cover the soil and protect it from erosion. Most European crops do not protect the soil as well because they come from European agricultural ecosystems that evolved under very different topographic and climatic conditions. European crops are sustainable in their places of origin, but they are not sustainable on tropical mountain hillsides where land with these crops can lose so much of its topsoil that agriculture is eventually no longer possible.

PORTABLE CAPITAL IN A FREE MARKET ECONOMY

Economic institutions frequently encourage people to use renewable resources in unsustainable ways. A common way to use forests on a sustainable basis is to cut a small percentage of the trees each year. If too many trees are cut, the tree population will decline and the trees will eventually disappear. The percentage of a forest's trees that can be cut on a sustainable basis each year depends upon the growth rate of trees. If the trees grow fast, a larger percentage can be cut each year. A typical growth rate for temperate forests is 5 per cent each year; the quantity of wood in the forest increases by 5 per cent during a year. To use a forest on a sustainable basis, no more than 5 per cent of the wood should be cut each year.

Imagine that you own 10 hectares of forest. You have two choices. You can cut 5 per cent of the wood each year for a sustainable harvest. Or you can cut all of the trees as soon as possible, sell the wood and invest the money in another business. If you invest the money in another business, return on the investment will be 10 per cent per year. However, if you cut all of the trees, it will be at least 40 years before there are mature trees on your land that will provide more timber. Which way will enable you to obtain the most money over the long term:

- Harvest the forest sustainably?
- Cut all of the trees and invest the money from the timber in another business?

The second choice provides the most long-term income. This example illustrates the fundamental conflict between the profit motive and sustainable use of natural resources. Our modern economic system has a strong effect on the way renewable natural resources are used because capital is 'portable'. Capital is portable because money is easily moved from one business enterprise to another. If decisions about the use of renewable natural resources are based exclusively on profits, even long-term profits, renewable natural resources will be used on a sustainable basis only if their biological growth rate is greater than the expected growth rate of alternative investments. Because the growth rate of the world economy today is greater than the biological growth rate of most renewable natural resources, there are powerful economic incentives not to use renewable natural resources on a sustainable basis. If people accept the rules of the game in a free market economy, it is rational to use renewable resources unsustainably whenever biological production fails to compete with alternative forms of investment.

tragedy of
the commons

TRAGEDY OF THE COMMONS

Commons means common property resource, a resource that is shared by many people. The atmosphere, oceans, lakes and rivers are commons that provide natural resources and absorb pollution. Forests, grazing lands and irrigation water may also be common property resources. Many commons are the property of no one in particular. Such commons typically have 'open access'; they can be used by anyone to any extent. Open-access commons are vulnerable to overexploitation because no one is responsible for controlling the intensity of their use.

Overexploitation under these circumstances is known as the **tragedy of the commons**. What is best for each individual is not best for all resource users together. For example, the Earth's atmosphere is a common property resource that is polluted by automobile exhaust. Air pollution from a single automobile is of little consequence, but pollution from all of the automobiles in a crowded city can create a serious health hazard. Carbon dioxide from automobile emissions is contributing to the global warming that is dramatically changing the global ecosystem.

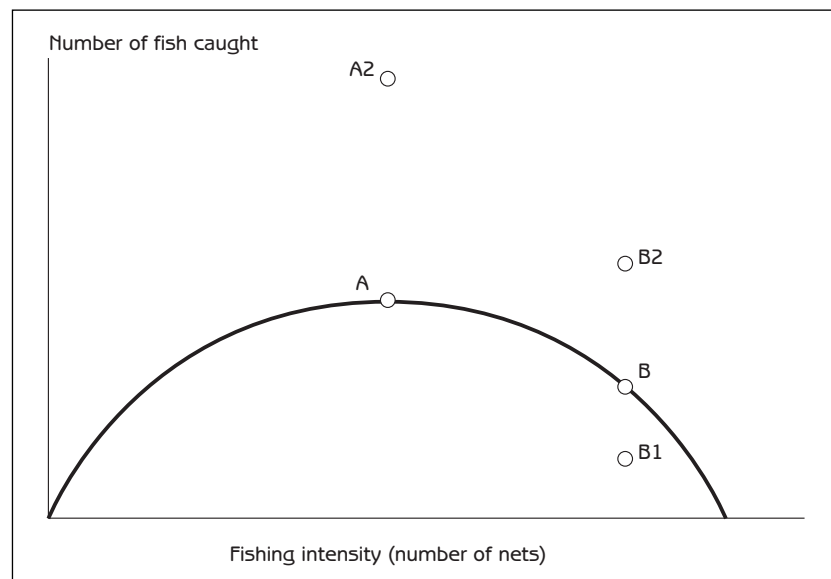
tragedy of the commons

Overfishing illustrates how tragedy of the commons is a consequence of 'rational' decisions by individual resource users to get as much of the resource as possible. It is best for sustainable fishing if all fishermen limit the number of nets that they use to no more than the optimum, as in Figure 10.3 (point A). Too many nets will reduce the fish population to such an extent that everyone catches less fish (point B). Tragedy of the commons occurs because the nets of one fisherman cannot catch enough fish to have a noticeable negative effect on the fish population. Each fisherman knows that he will catch more fish if he uses more nets, regardless of the number of nets that other fishermen are using. For one fisherman, twice as many nets catch twice as many fish (A_2) because the nets of one fisherman cannot catch enough fish to have an effect on fish stocks. However, if all fishermen use more nets, they will catch so many fish that the fish population is reduced, and the long-term fish catch will decline (point B). Tragedy of the commons can cause fishermen to use more and more nets until fish stocks disappear, because even when overfishing is severe, each individual fisherman catches the most fish by using more nets than other fishermen (B_2). A fisherman who uses a smaller number of nets, when other fishermen are overfishing, is punished by catching almost no fish at all (B_1).

Tragedy of the commons is rational for individuals, but it is not rational for society. Preventing tragedy of the commons is almost impossible with open access to resources, but it can be prevented if a resource has clear ownership with the owner (or owners) controlling who uses the resource and how it is used. This is known

Figure 10.3
The response of fish catches to fishing intensity as an example of tragedy of the commons

- A All fishermen use fewer nets to achieve a high sustainable catch.
- B All fishermen use more nets. Everyone catches less because overfishing reduces the fish population.
- A_2 One fisherman uses twice as many nets when all the other fishermen use fewer nets for a high sustainable catch.
- B_1 One fisherman uses fewer nets when other fishermen are overfishing.
- B_2 One fisherman uses twice as many nets as fishermen who are overfishing.



as closed access. Closed-access resources can also be overexploited, but overexploitation of these resources can be prevented if resource owners have **social institutions**, established rules for behaviour in a community, that give them the power to make sure everyone uses the resource in a sustainable manner. The next chapter will describe social institutions that prevent tragedy of the commons from occurring.

LARGE INPUTS TO AGRICULTURAL AND URBAN ECOSYSTEMS

large
inputs

People create agricultural and urban ecosystems by using energy inputs of materials, energy and information to modify the structure of ecosystems so that they function in ways that better serve human needs. In the past, energy inputs came from human and animal labour. Today, most of the energy comes from fossil fuels.

There is an important relationship between inputs and sustainability: agricultural and urban ecosystems are less sustainable over the long term if large quantities of human inputs are required to keep the ecosystem functioning in the way that people want. This is because it is difficult to ensure that large inputs can be provided on a reliable basis for a long time.

The experience of ancient Middle Eastern civilizations provides a good example. Civilization in the Middle East has always depended upon irrigation agriculture because the arid climate seriously limits natural biological production. Cities such as Babylon were able to develop along the rivers of Mesopotamia because water from rivers was used to create agricultural ecosystems that produced enough food to support cities. The cities lasted for centuries, but eventually they collapsed and were buried under desert sands because their agriculture collapsed.

The reasons for agricultural breakdown in these ancient civilizations were varied and complex, but one common explanation was the failure to maintain the ditches that transported irrigation water from the river to farm fields. River water contains sediment (eroded soil suspended in the water), which settles to the bottom of irrigation ditches as water travels along the ditches. If sediment is not removed from a ditch, it continues to accumulate on the bottom until the ditch is so full of sediment that it can no longer transport water. In ancient times, human and animal labour were used to remove sediment from the

ditches. Today in industrialized nations, machines do the job with petroleum energy.

Middle Eastern civilizations used large quantities of wood for the construction of their cities. The wood came from nearby mountains, as did the water in the rivers. After hundreds of years, deforestation and grazing by goats destroyed most of the vegetation that covered the mountain soil and protected it from erosion. Sediment increased in the rivers, the quantity of sediment that settled in irrigation ditches increased correspondingly, and more human and animal energy inputs were required to remove sediment from ditches. Before deforestation, nature performed the work of keeping sediment out of irrigation ditches by providing sediment-free water. After deforestation, the work shifted to people. Eventually, there was more sediment in the ditches than people could remove, particularly when the labour supply was reduced by demands from other sectors of the complex society, including emergencies such as war. The ditches filled with so much sediment that it was not possible to channel river water to the fields, irrigation agriculture collapsed and so did the civilization. Where this occurred, agriculture ecosystems were unsustainable because the social systems were not able to continue providing energy inputs large enough to maintain them.

URBANIZATION AND ALIENATION FROM NATURE

The inborn need for humans to learn about nature is apparent when we observe the curiosity that children have for nature and the intensity with which they explore their natural environment during casual play. Childhood experience generates an emotional attachment to nature and a concrete and intimate knowledge of nature in the locale where a child lives. This emotional need for nature has been termed **biophilia**. The childhood imprinting process, so essential to the full development of biophilia, appears to be a basic part of the human psyche, but it can only happen if a child has access to nature – an opportunity denied to children who live in cities with no natural ecosystems within their reach. The result may be adults who lack the emotional attachment to nature and knowledge of nature that is necessary for sustainable interaction. It may be that no amount of international treaties, government planning and regulations, or even environmental instruction in the classroom, will be sufficient if people lack the

love and respect for nature to compel them to conduct their everyday business in ways that do not destroy their environmental support system.

The potential significance of direct contact with nature during childhood can be appreciated by imagining a futuristic society in which children are separated from their families at birth and raised in dormitories. Even if the children receive daily indoctrination in school about love and respect for parents, the way that they relate to parents as adults will be fundamentally different from adults who were held in their mothers' arms and experienced the full richness of family interactions throughout childhood. Similarly, a concern for the environment that comes only from school may lack the substance and depth that is necessary for a society to be ecologically sustainable.

complex
societies

THE RISE AND FALL OF COMPLEX SOCIETIES

A prominent characteristic of urban social systems is their social complexity characterized by extensive differentiation and specialization of social roles and elaborate organization of human activities. The role of social complexity in the growth and decline of cities can be seen in past civilizations, such as those in Mesopotamia, Egypt and Greece, as well as Mayan and Pueblo Indian civilizations of the Western Hemisphere. These civilizations experienced cycles of growth and decline that extended over centuries. The European empires experienced a similar process of growth and decline during the past 400 years.

The growth of complex societies

Urban ecosystems were primarily simple villages before the Agricultural Revolution. They were small and nearly independent of one another. Village social systems were egalitarian; nearly everyone was of equal status. There was division of labour by sex, but the number of distinct social roles was small. Almost everyone was a jack of all trades, doing whatever was necessary for a **subsistence** way of life. Most families produced their own food, made their own clothing and built their own houses. There was some specialization, but the simplest societies had as few as 25 different occupational roles. Though very sophisticated in other ways, all human social

systems before the Agricultural Revolution were relatively simple with regard to occupational roles. Pre-industrial societies in more isolated parts of the world still exist in this way.

The formation of cities as larger and more complex urban ecosystems is only possible when a society's agricultural ecosystems are productive enough to supply a surplus of food beyond the needs of the families who produce the food. The extra food allows city people to specialize in a variety of non-farming occupations – a **division of labour** that forms the core of social complexity. It is typical for modern society to have more than 10,000 different occupational roles in a single large city.

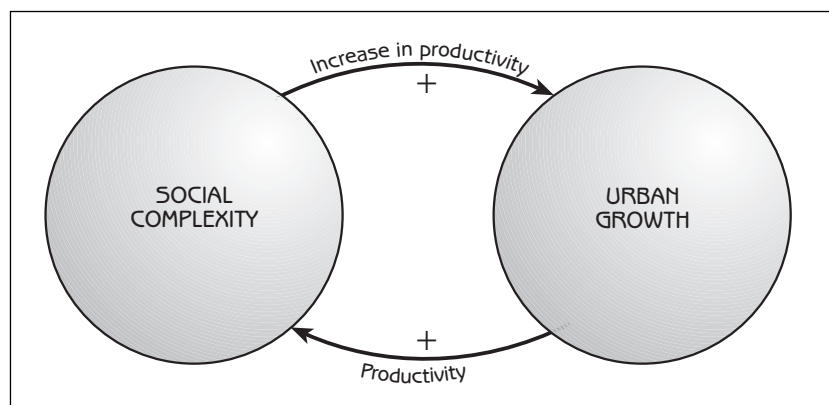
complex
societies

Many of these occupations work in support of the society's productive activities to make them more effective. There is also the opportunity for innovators and artists to flourish. Potters and poets, engineers and scientists, teachers and priests all add to the richness of life. However, such extensive division of labour requires that large amounts of time and energy are spent in processing information, communicating, distributing goods and keeping track of property ownership and the exchange of goods and services. This is true today and it was equally so for civilizations in the past. Records from ancient civilizations commonly feature voluminous and detailed accounts of commercial exchanges and inventories of goods.

Division of labour in complex societies is not only occupational. There is social stratification with regard to power and wealth, and there is a hierarchical authority structure in the form of government.

Past civilizations were associated with cities, or clusters of cities, that grew in size and complexity for centuries, extending their influence over a larger and larger area, their **zone of influence**. This expansion was not always peaceful and often involved the exploitation of conquered people and their resources.

Figure 10.4
Positive feedback
loop between
social complexity
and the growth
of cities



Once started, it is typical for complexity to continue increasing. This leads to further growth and expansion, which creates a need for additional productivity by means of greater complexity (see Figure 10.4).

Growth and complexity thus form a **positive feedback** loop that makes them increase exponentially. A city can eventually grow to dominate such a large area, and it can be so complex and with so many social controls that it appears the power and wealth of the city will continue forever. This is the *equilibrium* stage of the complex system cycle.

The decline of complex societies

complex
societies

Eventually the social system of a complex civilization becomes too complex to continue functioning effectively. When social complexity is greater than the optimum (see Figure 10.5), more complexity can lead to less productivity because of the following:

- Many of the benefits from social complexity have **diminishing returns**. Once a certain level of technology and organization of human activities is reached, more intensive technology and organization do not produce more results (see the benefits curve in Figure 10.5).
- Social complexity has a substantial cost in terms of the energy and effort needed to organize and maintain it (see the cost curve in Figure 10.5).
- Costs continue to increase, even when there are no additional benefits from greater complexity, so productivity (which equals benefits minus costs) declines (see the productivity curve in Figure 10.5).
- Society develops cultural values that encourage additional complexity. This can lead to a random proliferation of complexity that lacks the structure to contribute to functionality in the system as a whole.

As a result, the productivity of the society, as well as its standard of living, decline. It might appear logical for a social system with too much complexity to reduce its complexity to the optimum. However, social systems usually become even more elaborate because societies have evolved a cultural belief that more complexity is the best way in which to deal with problems.

By the time a society has exceeded optimum complexity, it is common for it to start experiencing serious environmental problems, and *dissolution* begins. Damage due to excessive

complex
societies

demands on the ecosystem that has been building up slowly for centuries finally leads to a decline in agricultural productivity and a shortage of food and other renewable natural resources (see Figure 10.6). The society no longer has the surplus food and other resources that it needs to support a large urban population and maintain food reserves. As reserves of food and other essential resources diminish, the society loses its **resilience** to cope with further decline. (Resilience is discussed in Chapter 11.)

When the standard of living declines, communities in the city's zone of influence become dissatisfied and try to break their ties with the city (see Figure 10.6). In response, political authorities may rely on military force to compel surrounding communities to continue supporting the city. Political authorities may also channel more resources into projects such as monuments or elaborate ceremonies to glorify the image of the civilization. Because military expenditures and glorification projects are expensive but do not increase productivity, the standard of living declines even further. Greater demands are placed on people and the ecosystem in the surrounding area to increase production. There is more environmental damage, agricultural productivity declines, the standard of living declines and people are more dissatisfied. These

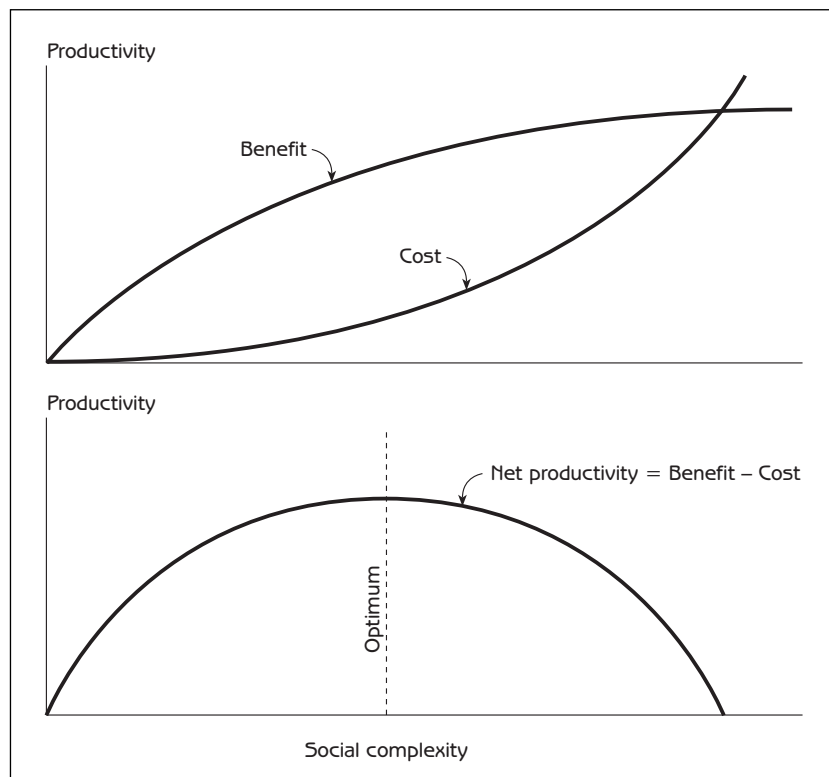


Figure 10.5
Benefits and
costs of social
complexity

downward positive feedback loops may eventually cause the city to be abandoned. The city's inhabitants migrate to places where opportunities are better (*reorganization*), and a new city begins to grow somewhere else.

The rise and fall of complex societies is not just a story of ancient civilizations. Contemporary urban ecosystems experience complex system cycles of growth and decline on a spatial scale that extends from neighbourhoods and small cities to large cities, metropolitan areas and entire civilizations. A neighbourhood grows as people or commercial activities move to it from other neighbourhoods. A few decades later, the same neighbourhood may decline as people and commercial activities move to other competing neighbourhoods. The same thing happens with whole cities on a longer time scale.

Because this example of urban growth, collapse and reorganization is as old as human civilization, the modern world's story of population explosion and an expanding global economy is not so new, except in one very important way. Until recently, the growth and decline of urban ecosystems have been on local or regional scales. When cities or regional civilizations collapsed, people moved to new areas. Now, with global transportation and communications and a global economy, human social systems are becoming a single global social system, and the Earth's ecosystems are becoming strongly connected through human activities. For the first time in history, the growth of human population and social complexity is happening simultaneously in almost every urban ecosystem and social system over the entire planet. While growth, collapse and migration were local or regional in the past, there is now a possibility of global collapse with nowhere to move.

complex
societies

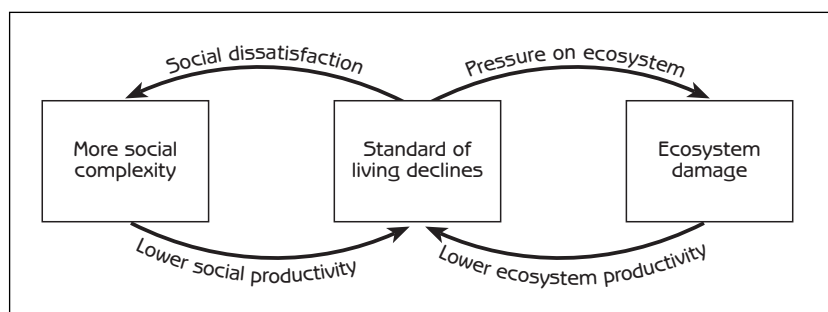


Figure 10.6
Positive feedback loops for the dissolution phase in the rise and fall of complex societies

WISHFUL THINKING AND THE PRECAUTIONARY PRINCIPLE

precautionary principle

With the increase in environmental awareness during recent years, many governments are taking measures to prevent overexploitation and depletion of natural resources. Ironically, some renewable resources have collapsed since the effort to protect them began. In part, this has happened because population growth and economic growth have made greater demands on ecosystem services than before. However, many of the attempts to protect renewable resources from overexploitation have been unsuccessful because people were not realistic about the limits of the resources. I call this wishful thinking.

The history of ocean fisheries during the past two decades illustrates the hazards of wishful thinking. About 20 years ago, coastal nations declared their ownership of the ocean and its resources to a distance of 320 kilometres from their shores. These areas are known as extended **economic zones** and cover the fish, petroleum and minerals in the zones. Governments formulated management plans and regulations to control the quantity and kind of fishing by their own fishermen and foreign fishermen in the extended economic zones. A common practice was to manage for 'maximum sustainable yield' – the largest fish catch possible on a long-term basis. Management plans were generally based on advice from scientists and representation from the fishing industry. In the years that followed, a number of commercially valuable fish populations collapsed despite the effort to protect them. Many of the management plans were not completely successful. Why did this happen?

Firstly, scientists had to work with imprecise information, and often management plans did not allow for the possibility that the estimates of fish production and the capacity of fish populations to withstand fishing might be wrong. Secondly, management plans did not take into account the way that fish populations change from year to year because of natural fluctuations in the physical and biological conditions of the oceans (ie, natural changes in ocean ecosystem state). During some years physical conditions and the supply of food in the oceans are more favourable for new generations of fish stocks; during other years young fish have difficulty surviving. As a result, fish populations can withstand more fishing during some years, while during other years stocks are seriously depleted by the same amount of fishing. Thirdly, improvements in fishing technology can increase the number of fish that fishermen catch. When this happens, it is necessary to

revise the regulations to ensure that fish catches stay within sustainable bounds.

Governments accepted optimistic but risky plans because the fishing industry wanted to catch as many fish as possible. In their initial forms, management plans frequently allowed fishermen to catch the same number of fish every year, whether it was a good year for fish or a bad year. Fish populations collapsed when a plan overestimated the sustainable catch, when the allowable catch was too close to the limit or when regulations were not revised when conditions changed. Fish populations could not withstand fishing close to the limit during several years in a row of unusually low fish production (see Figure 10.7). The most spectacular collapse was the North Atlantic cod fishery, which had provided livelihoods for thousands of fishermen for several centuries. The changes may not be reversible. Commercially valuable fish populations may not return, even if fishing is reduced.

The lesson from this story is that ecosystem services can be used on a truly sustainable basis only if the intensity of use is substantially less than the apparent maximum. This is the **precautionary principle**. Pushing ecosystems to the limit is risky because of imprecise estimation of the limits, as well as fluctuations in the capacity of ecosystems to provide the services. If people push the ecosystem state too close to the boundary between stability domains by using an ecosystem service too intensely, natural ecosystem fluctuations may push the ecosystem into a stability domain with diminished ecosystem services (see Figure 10.7).

The precautionary principle has become a major instrument for elaborating environmental policies. It reflects common sense. A Japanese Zen proverb says: 'The care with which a blind man crosses a log bridge is a good example of how we should live our lives.' The probing of a blind man is similar to the assessment

**precautionary
principle**

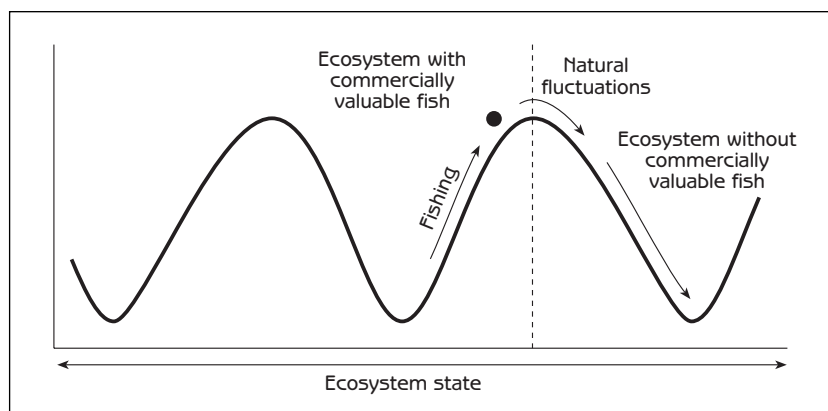


Figure 10.7
Change from one stability domain to another when fishing is too close to the boundary between stability domains in a fisheries ecosystem with natural climatic fluctuations

154 *Human Ecology***precautionary
principle**

process in adaptive development described in Chapter 11. While the precautionary principle can contribute to ecological health and sustainability over a broad spectrum of human–ecosystem interaction, its implementation is far from straightforward in a social system strongly committed to maximum possible exploitation of ecosystem resources and services. The simple wisdom of the precautionary principle is counter to the prevailing ethos of the modern global economy, which rewards entrepreneurial boldness and portrays confidence in economic growth as an ultimate virtue.

Practical implementation of the precautionary principle cannot be separated from short-term tradeoffs between ecological sustainability and other social concerns. A recent episode involving commercial fishing in Hawaii illustrates how people can have radically different opinions about where to draw the line (see Chapter 1). Several conservation organizations have initiated legal action to compel the United States government to close down long line fishing in areas where long lines kill marine mammals, seabirds, turtles and fish that are not an intended part of the catch. The consequences of this legal action are numerous. Closing long line fisheries will reduce employment for fishermen and reduce the supply of fish for consumers. Whether the fisheries are closed or not, dealing with this issue entails government expenses for litigation and research that can divert funds from other government programmes benefiting the same species (for example, protecting beaches where marine turtles lay their eggs). An answer to the question: ‘To what extent is long line fishing harming these species, and how will closing the fisheries really benefit them?’ is crucial to identifying appropriate restrictions on long line fisheries. However, scientific information is limited and interpretation of this information by legal adversaries can be biased. Some people believe that long lines kill few turtles and that modifications in fishing methods can protect other marine species. They prefer to modify fishing methods and see what happens, before resorting to more drastic measures such as closing an entire fishery. Other individuals, who have a lower tolerance for even potential damage to the marine ecosystem, believe that long line fishing should simply stop.

This example has far-reaching implications. Today’s modern social system is vulnerable to wishful thinking not only about the magnitude of demands that ecosystems can sustain but also about the ability of modern science and technology to manipulate ecosystems to satisfy these demands. We do not have enough scientific knowledge about ecosystems to know precisely how far we can change them without risking collapse. Even with more

scientific information and more powerful computers to process the information, ecosystems are so complex that we may never know precisely in advance what the consequences of our actions will be. Moreover, micromanagement of the vast ecosystems on which we depend for survival is a practical impossibility. There are not enough scientists – and there is not enough human labour and fossil fuel energy – to fix all of the problems created by misusing ecosystems, and then to fix all of the new problems created by ecological intervention.

THINGS TO THINK ABOUT

things to
think about

1. Look at the example of portable capital in a free market economy. The first alternative, harvesting the forest sustainably, provides a long-term supply of timber. The second alternative, cutting all of the trees and investing the money in another business, does not. Do you understand why the second alternative provides the highest long-term income? The second alternative explains why portable capital makes the unsustainable use of biological resources with slow growth rates a rational option from a business perspective. Think of concrete examples that illustrate unsustainable human–ecosystem interaction because of portable capital.
2. Think about sources of unsustainable human–ecosystem interactions in the social system in which you live. Think firstly about the local level, and then think about the national and international levels. Consider concrete examples that demonstrate how human–ecosystem interactions can be less sustainable because of:
 - human migration;
 - technology;
 - tragedy of the commons;
 - the economic system;
 - urbanization.
3. List specific benefits of social complexity at different levels of your social system (local community, nation, global society). What are some of the costs of social complexity at these different levels of social organization? Where do you think your social system is located on the curve in Figure 10.5? Is its social complexity near, less than or greater than the optimum?
4. Childhood contact with nature seems to be important in order to develop the knowledge and judgement that people need for sustainable interaction with ecosystems as adults. What kind of contact did you have with nature as a child? Talk to people with childhood experiences similar to your own, as well as to people with very different experiences. How did your childhood experiences influence your attitudes and actions toward ecosystems as an adult?
5. Identify ways in which agricultural and urban ecosystems in your region are dependent upon large quantities of energy or other resources,

156 *Human Ecology*

things to
think about

making them vulnerable to price increases or shortfalls in supply. What can be done to reduce this vulnerability?

6. What are the ways in which your local and national community leaders (business, government, etc) exhibit wishful thinking concerning the demands that the community or nation can safely place on ecosystems? How do their attitudes and actions compare with the precautionary principle? Why do community leaders think the way they do? Do most citizens have attitudes and actions similar to their community leaders? If there is a diversity of attitudes and actions, what are the rationales for different approaches? Are there particular ways in which you think that people should pay more attention to the precautionary principle?